# Machine Code Following



## 1. Course Content

- Run example programs, the car will recognize detected machine codes and frame them, then the car will continuously follow the machine code to keep it in the center of the screen, and maintain a distance of 0.4 meters from the recognized machine code. Press the `q/Q` key to exit the program.

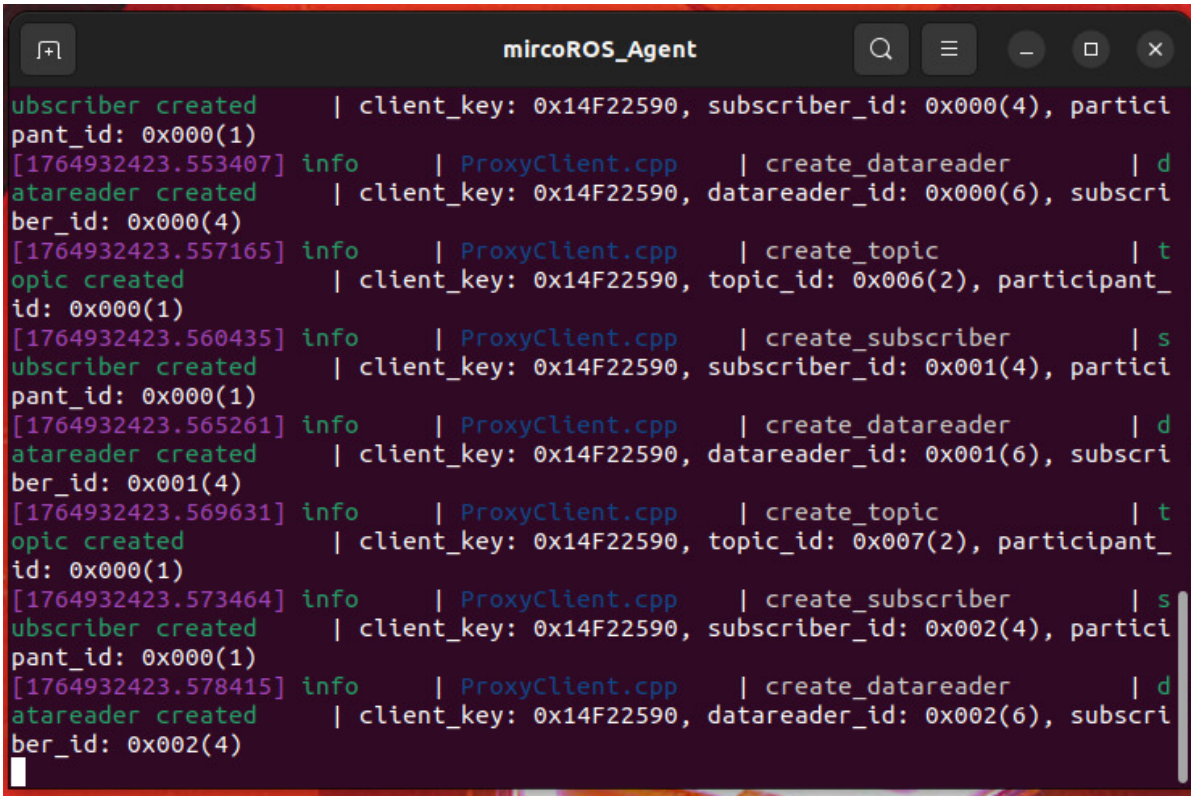
## 2. Start Agent

**Note: If already started, no need to restart**

Enter command in the car terminal:

```
start_agent
```

The terminal prints the following information, indicating successful connection




## 3. Run Case

**For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,**

```
bringup_m3
```

**Open a terminal inside the container,**

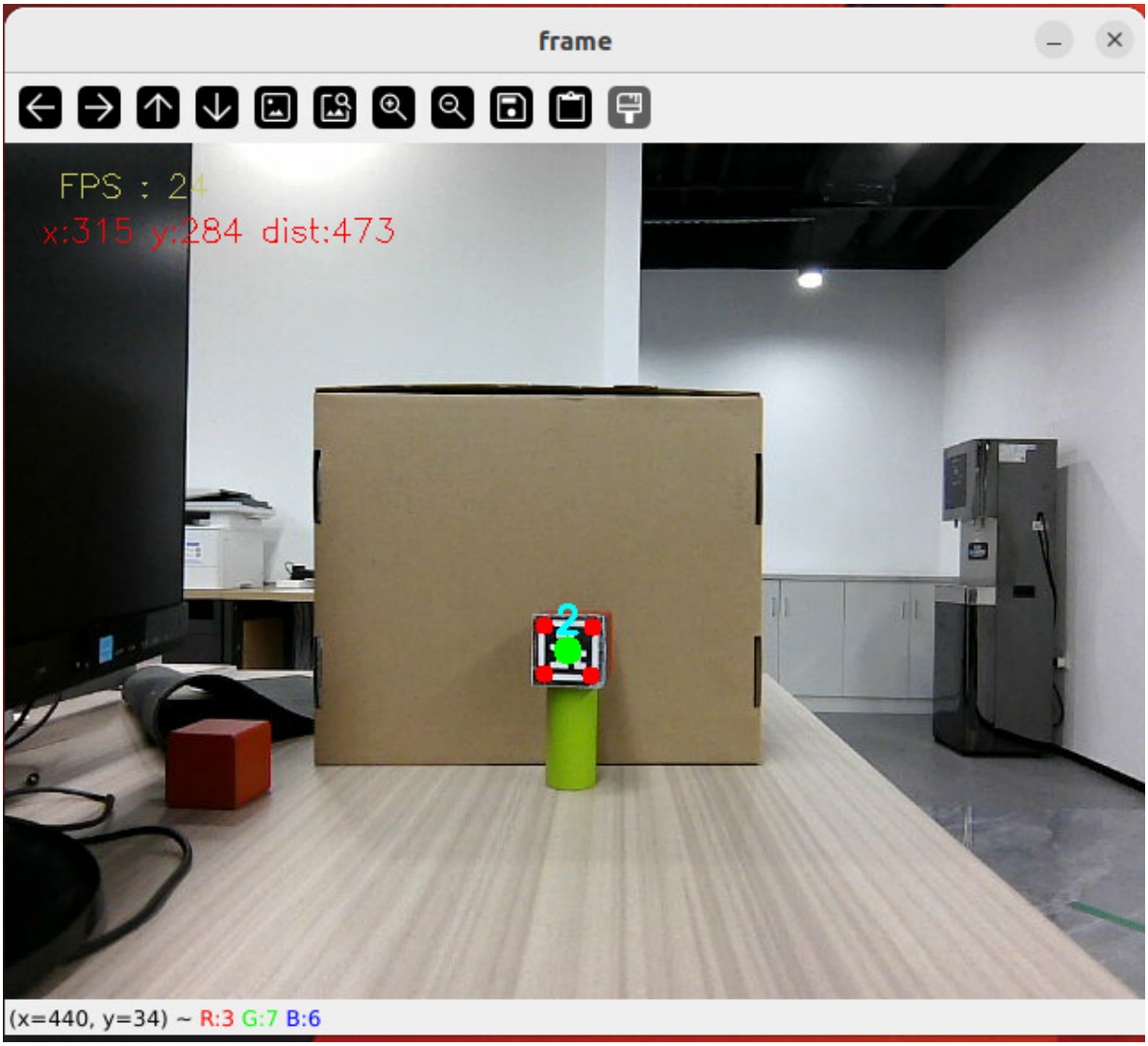
```
exec_m3
```

- Enter command in terminal:

```
ros2 launch m3_bringup depthcam_demo.launch.py demo:=apriltagTracker
```

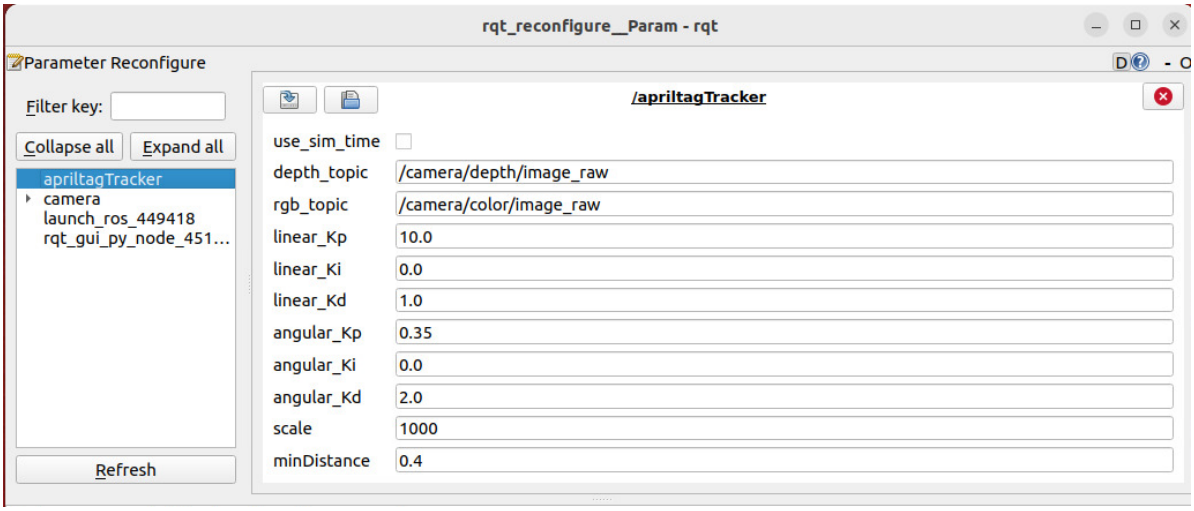
- After the program starts, when a machine code is detected, it will automatically frame the machine code area, and the car will start following.

> [!TIP]
>
> If the car runs with poor stuttering effect, you can replace it with a QR code photo that can better obtain depth information. This is because if the QR code depth information is 0, the car will stop



## 3.1 Dynamic Parameter Adjustment

```
ros2 run rqt_reconfigure rqt_reconfigure
```




- After modifying parameters, click on the blank area of the GUI to write parameter values. Note that it will only take effect for the current startup. For permanent effect, you need to modify the parameters in the source code.

**Parameter analysis:**

【linear_Kp】, 【linear_Ki】, 【linear_Kd】: Linear speed PID control during car following process.

【angular_Kp】, 【angular_Ki】, 【angular_Kd】: Angular speed PID control during car following process.

【minDistance】: Following distance, always maintain this distance.

# 4. Source Code Analysis

`apriltagFollow.py` source code path:

```
~/yahboomM3_ws/code_ws/src/m3_common_demo/m3_common_demo/depthcamera/apriltagFollow.py
```

- This program mainly has the following functions:
- Subscribe to depth camera topics to get topic images;
- Use machine code dt_apriltags library to detect recognized machine codes;
- Process images to get machine code screen center coordinates and depth distance information.
- Calculate distance PID and center coordinate PID to get car forward speed distance and turning angle.

Some core code is as follows,

```
#AprilTag detection and processing
tags = self.at_detector.detect(
    cv.cvtColor(result_image, cv.COLOR_BGR2GRAY),  # Convert to grayscale
    False,
    None,
    0.025)  # Decoding threshold parameter
```

```
tags = sorted(tags, key=lambda tag: tag.tag_id)  # Sort by ID
# Draw labels: red corner point, green center point
result_image = draw_tags(result_image, tags, corners_color=(0,0,255),
center_color=(0,255,0))
#Machine code detection and processing
if len(tags) > 0:
    tag = tags[0]  # Only process the first detected label
    self.Center_x, self.Center_y = tag.center  # Get tag center
    corners = tag.corners.reshape((-1, 2))
    # Calculate bounding boxes
    x, y, w, h = cv.boundingRect(corners.astype(np.int32))
    self.Center_r = w * h // 100  # Area as a radius indicator
#Distance calculation
points = [
    (int(self.Center_y), int(self.Center_x)),
    (int(self.Center_y+1), int(self.Center_x+1)),
    (int(self.Center_y-1), int(self.Center_x-1))
]

valid_depths = []
for y, x in points:
    # Get depth value (note: OpenCV is row, col is y, x)
    depth_val = depth_image_info[y][x]
    if depth_val != 0:  # Filter invalid depth
        valid_depths.append(depth_val)

dist = int(sum(valid_depths)/len(valid_depths)) if valid_depths else 0
self.dist = dist  # Store current distance
#Control execution
if dist > 0.2:  # Effective distance threshold (0.2 meters)
    self.execute(self.Center_x, dist)  # Execution control

# Calculate the movement speed and turning angle using the PID algorithm based on
the x and y values
def execute(self, rgb_img, action):
    #PID calculation deviation
    linear_x = self.linear_pid.compute(dist, self.minDist)
    angular_z = self.angular_pid.compute(point_x,320)
    # The car is in the parking area and the speed is 0
    if abs(dist - self.minDist) < 30: linear_x = 0
    if abs(point_x - 320.0) < 30: angular_z = 0
    twist = Twist()
    ...
    # Publish the calculated speed information
    self.pub_cmdVel.publish(twist)
```